Tectonophysics 463 (2009) 69–76

Contents lists available at ScienceDirect

Tectonophysics

journal homepage: www.elsevier.com/locate/tecto

New structural implications for the central Sahara (Algeria), from the revisited Upper Carboniferous "Hassi Bachir" Formation: Paleomagnetic constraints

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ARTICLE INFO

Article history:
Received 14 May 2008
Received in revised form 9 September 2008
Accepted 10 September 2008
Available online 19 September 2008

Keywords:
Paleomagnetism
Upper Carboniferous
Tilt test
Tectonics
Central Sahara

ABSTRACT

Paleomagnetic investigations of the folded Upper Namurian–Lower Moscovian "Hassi Bachir" Formation cropping out in the "Ahnet" basin (Central Sahara, Algeria) yield two magnetic components. A pre-folding primary magnetization (D=136.1°, I=22.0°, k=217, α_{95}=2.6°) enables us to define a paleomagnetic pole (32.8°S, 55.7°E, K=328 and A_{95}=2.0°) which better constrains a paleopole that was determined by Daly and Irving [Daly, L., Irving, E., 1983. Paléomagnétisme des roches carbonifères du Sahara central; analyse des aimantations juxtaposées; configurations de la Pangée. Ann. Geophys. 1, 207–216] for the same formation. A secondary component consists in a synfolding remagnetization and shows that post-Permian tectonics account for at least about half of the total folding in the studied area. This indicates that Mesozoic folding noted 150 km to the West in the Reggane basin [Smith, B., Derder, M.E.M., Henry, B., Bayou, B., Amenna, M., Djellit, H., Yelles, A.K., Garces, M., Beamud, E., Callot, J.P., Eschard, R., Chambers, A., Aifa, T., Ait Ouali, R., Gandriche, H., 2006. Relative importance of the Hercynian and post-Jurassic tectonic phases in the Saharan platform: a palaeomagnetic study of Jurassic sills in the Reggane basin (Algeria). Geophys. J. Int. 167, 380–396] is not local and affected at least the entire north-western part of the Hoggar area. This reconfirms that the folding of the Paleozoic cover in the Sahara platform should not be restricted to the Hercynian orogeny.

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1. Introduction

During the last two decades, numerous paleomagnetic studies from the Algerian Saharan craton were conducted in order to improve the poorly determined segments of the Apparent Polar Wander Path (APWP) for Africa. These studies have demonstrated the importance of widespread secondary magnetizations, mostly remagnetizations of Permian and Cenozoic ages (Henry et al., 2004a). Furthermore, the determination of several new paleomagnetic poles has greatly improved the African APWP from the Lower Carboniferous to Dogger times (see Mérabet et al., 2005 and references in; Derder et al., 2006 and references in). Although these new data are reliable and satisfy modern criteria of paleomagnetic studies, the age of magnetization, in most cases, has not been well constrained by paleomagnetic tests, particularly fold tests (which are actually s.s. tilt tests — see Borradaile, 1997). This underlines the importance of investigations in folded areas over cratonic regions.

Another aspect of these earlier studies was that the folding of the Paleozoic sedimentary cover in the Central Sahara platform was considered in general to occur mainly in the Hercynian (e.g. Conrad, 1984; Fabre, 1988; Haddoum et al., 2001). However a recent paleomagnetic study demonstrated the relative importance of the Mesozoic folding phase in the eastern part of the "Reggane" basin (Fig. 1) (Smith et al., 2006). In the neighboring "Ahnet" basin (Fig. 1), Daly and Irving (1983) isolated both a primary magnetization and a Permian remagnetization (i.e. a pre-tilting magnetization and a magnetization acquired between the two folding phases) in the folded Carboniferous "Hassi Bachir" Formation. This formation provides an excellent opportunity to determine whether the Mesozoic folding was a local event, limited to the "Reggane" basin or whether it affected other Saharan basins as well. The implications of such a result could be of major importance, in particular for petroleum research. The Daly and Irving (1983) study was only based on 31 samples from 7 sites, without tilt test and without a complete demagnetization of the samples. For this reason we reinvestigate the "Hassi Bachir" Formation by extending the sampling to include sites covering a larger area and showing various dips in order to allow for a tilt test. We also conducted a preliminary paleomagnetic study on the underlying "Djebel Berga" limestone, which is attributed to the Lower Namurian. The main aim of this study is of structural interest; however a secondary goal is to improve the corresponding paleomagnetic data, still considered as a key reference pole for the African APWP.

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doi:10.1016/j.tecto.2008.09.012

2. Geological setting

The "Djebel Berga" area (~26° 40'N, 1° 50'E) is located in the "Ahnet" basin, central Sahara (Fig. 1). It corresponds to a structural basin with a NW–SE-trending axis and a flat base (Fig. 1). In its southeastern terminal curvature, the dip is towards the north-west with a low plunge (5° on average), while on its western and eastern narrow flanks, strata dip with a higher angle (20° on average), toward the north-east and the south-west respectively. In the central part of the basin, the beds become horizontal. The northern part of the basin is unconformably overlain by tabular Lower Cretaceous levels called "Continental Intercalaire" (Kilian, 1922). In this basin, two Upper Carboniferous series outcrop:

- The lower one (about 350 m thick) is the marine "Djebel Berga" limestones of late Visean–lower Namurian age—i.e. Serpukhovian, according to Conrad and Legrand Blain (1971). It includes at the base an alternation of limestones and sandstones, overlain by massive calcareous beds containing brachiopods (Gigantoproductus) and polyparies, interbedded with levels of fine sandstones and claystones (Conrad, 1973; Fabre, 1976).
- The upper series is the continental red "Hassi Bachir" Formation (about 250 m thick) of Upper Namurian–Lower Moscovian age (Conrad, 1973; Conrad and Lemosquet, 1984). It is mainly composed of red sandstones and argillites with ferruginous minerals, alternating with rare calcareous and dolomitic beds of marine or lacustrine origin (Conrad, 1973).

From a structural point of view, two main tectonic phases affected this region and folded the Paleozoic series (Conrad, 1981). The first one is of post-Moscovian age (Conrad, 1984). Some authors (Fabre, 1988; Haddoum et al., 2001) argue that it could be of Early Permian age. The second phase, possibly responsible for an important part of the total folding (Smith et al., 2006), occurred between the Lower Jurassic and the Early Cretaceous (Conrad, 1981, 1984). However, we observed the presence of slumps in the "Hassi Bachir" Formation on the south-western border of the basin (area with relatively high dip). These slumps are "sealed" by overlying levels, clearly indicating that

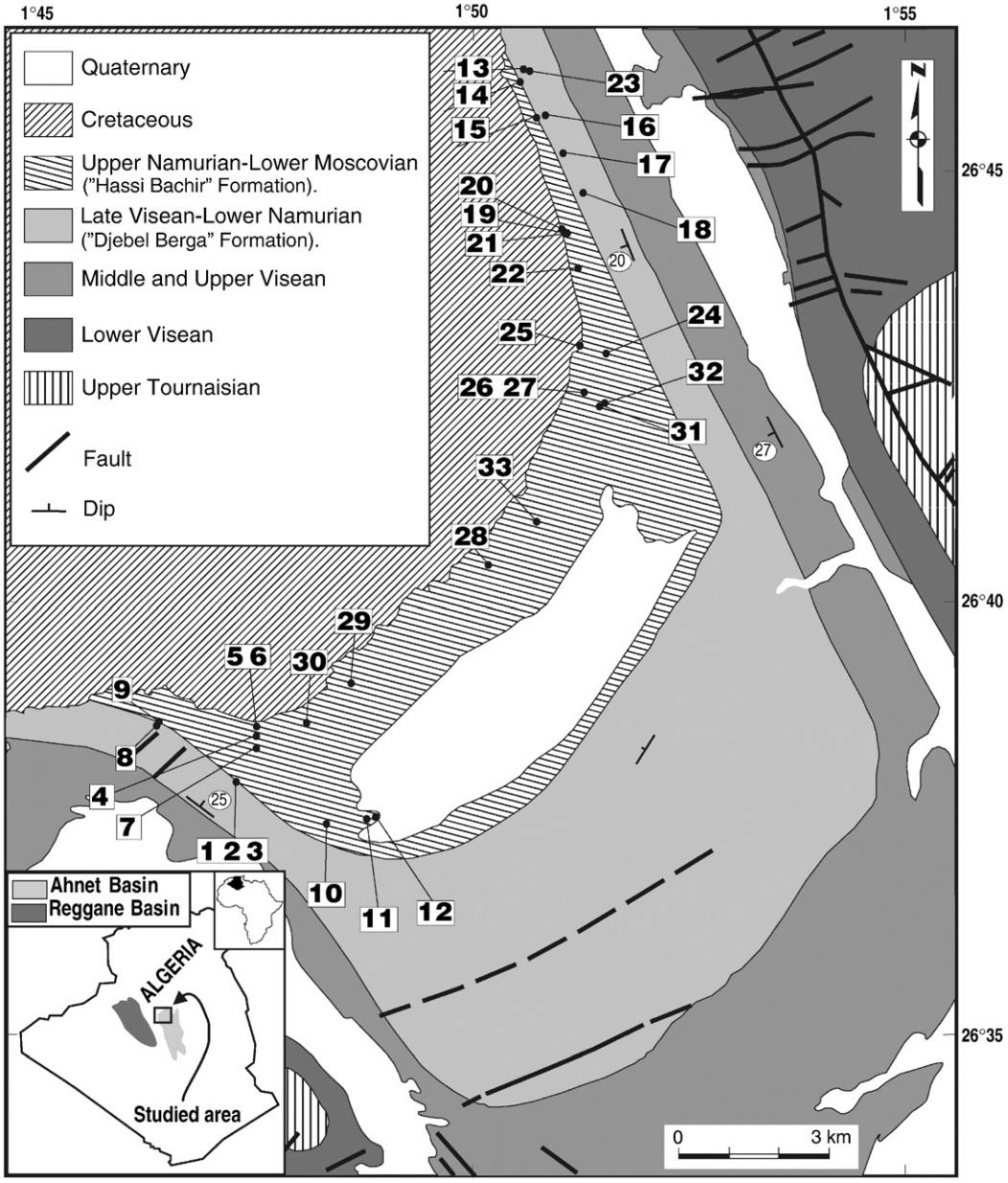


Fig. 1. Geological map of the "Djebel Berga" area and location of the sampling sites.

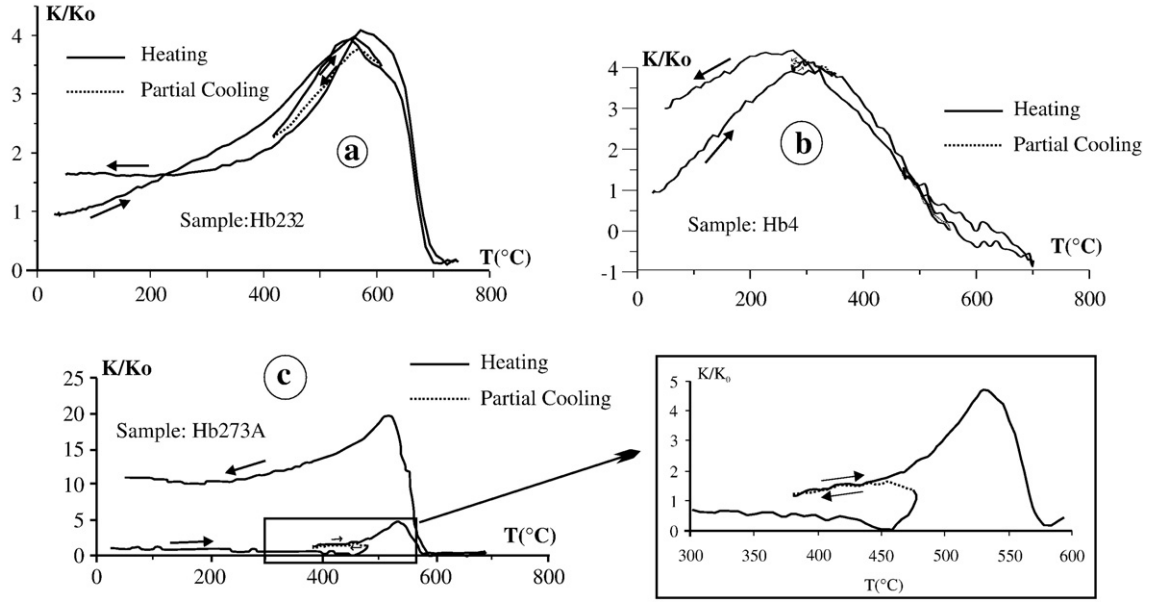


Fig. 2. Variation of the normalized susceptibility K/Ko as a function of the temperature *T* (thermomagnetic curves; dotted lines: partial cooling curves) in samples Hb232 (a), Hb4 (b), and Hb273A (c), showing in this latter case a strong mineralogical alteration during heating at temperatures higher than 400 °C.

significant subsidence occurred during deposition of the Upper Carboniferous formations. The synsedimentary deformation was probably concentrated along two NW–SE flexures located at the present NE and SW limits of the Upper Carboniferous formation.

3. Sampling and analytical procedure

The different facies of the "Hassi Bachir" Formation have been extensively sampled. The sampling covers the two borders of the basin where the bedding has a higher dip compared to the central part. A total of 307 oriented cores distributed over 28 sites (Fig. 1) have been collected. The underlying "Djebel Berga" Formation of late Visean–lower Namurian age has been also sampled. Unfortunately, due to bad outcrop conditions and unfavorable rock facies for paleomagnetism, only 60 oriented cores from 5 sites were collected and all from the north eastern border of the basin (Fig. 1).

One to three specimens were cut from each core in order to obtain additional specimens for pilot studies and rock magnetism analyses. Prior to any demagnetization analysis, the specimens were stored in a zero field for at least one month, in order to reduce possible viscous components. The Natural Remanent Magnetization (NRM) of the specimens was measured using a JR5 spinner magnetometer (AGICO, Brno). Several pilot specimens from each site were subjected to a stepwise alternating field (AF) demagnetization up to 100 mT or thermal (TH) demagnetization up to 670 °C in order to characterize the components of magnetization. Following the demagnetization process proposed by Daly and Irving (1983), some other specimens were subjected to combined AF and TH treatments.

A significant number of samples have very weak magnetizations and only 163 samples were suitable for demagnetization in this study, with 114 samples from the "Hassi Bachir" Formation and 49 from the "Djebel Berga" Formation.

The results obtained with the pilot studies showed that the AF and combined procedure did not allow complete demagnetization. Thermal demagnetization was then applied to the remaining specimens. It was performed using a cylindrical furnace with a fast heating level while forced air enhanced the cooling. In order to isolate correctly and identify the components of magnetization, numerous steps were performed with increments ranging from 100 °C at the lowest temperatures to 10 °C at the highest temperatures. Demagnetization data analysis was carried out using classical methods: the directions of the magnetization components were computed using principal component analysis (Kirschvink, 1980) and plotted on orthogonal vector plots (As and Zijderveld, 1958; Zijderveld, 1967); the remaining vectors and vectorial differences of the magnetization were plotted on equal area projections. The mean directions of the different components were computed using (Fisher, 1953) statistics and the bivariate form of Fisher statistics (Le Goff, 1990; Le Goff et al., 1992). For progressive untilting, we considered accompanying changes in the following statistical parameters for the orientation–

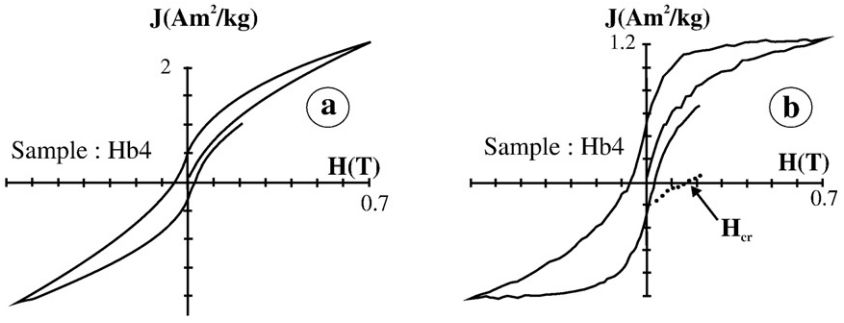


Fig. 3. Hysteresis loops before (a) and after correction of the paramagnetism in sample HB4.

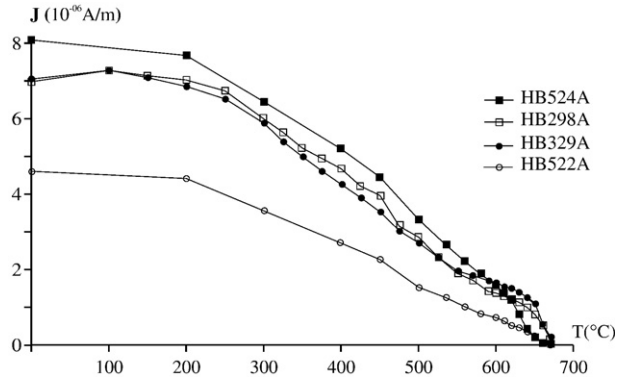


Fig. 4. Typical thermal demagnetization curves of different samples from "Hassi Bachir" Formation.

distribution of the paleomagnetic directions: the precision parameters k (Fisher, 1953) and the ellipticity (ratio ky/kx of the precision parameters for the short and long axes of the confidence ellipse — Le Goff, 1990; Le Goff et al., 1992). Fold tests of McFadden (1990) and Tauxe and Watson (1994) were performed.

4. Rock magnetism

Thermomagnetic curves (low-field magnetic susceptibility as a function of temperature) have been measured for representative samples by heating in air using the CS2–3 oven and the Kappabridge KLY3 (AGICO, Brno). Most samples show reversible magnetization during heating and cooling; therefore no chemical alteration occurs that could change the ferromagnetic mineralogy. During heating, the susceptibility first shows a large increase (Hopkinson effect) before a strong decrease up to 650–680 °C. This behavior is typical for hematite, which can be deduced to be the principal ferromagnetic mineral (Fig. 2a). Besides hematite, there is also a weak amount of magnetite as seen from the slight decrease in susceptibility at temperatures around 580 °C (Fig. 2b). Part of the samples, particularly from the "Djebel Berga" Formation, are affected by strong mineralogical alteration during thermal treatment at temperatures higher than 400 °C (Fig. 2c).

Hysteresis loops were obtained for small cores (cylindrical samples of about 3 cm^3) using a laboratory-made translation inductometer within an electromagnet capable of reaching 1.6 T. Saturation of magnetization is not reached (Fig. 3a) and the coercive forces are moderate to high (Hc ~ 76 mT, Hcr ~ 316 mT in the samples where the values can be reliably determined). This confirms that the principal carrier of magnetization is hematite. In most cases, the loops show a wasp-waisted shape (Fig. 3b), confirming the existence of at least two different magnetic carriers types (magnetite and hematite).

The thermal demagnetization curves (Fig. 4) show maximum unblocking temperatures higher than 620 °C in all samples, confirming hematite as principal magnetic carrier. In some samples, a weak inflection of the curve records another maximum unblocking temperature just below 580 °C, probably because magnetization is also carried partly by magnetite (samples HB298 and HB329 on Fig. 4).

5. Paleomagnetic results

5.1. The "Hassi Bachir" Formation

Two main components B and C are isolated during demagnetization in addition to a viscous component A (Fig. 5). The B component is defined between 200–250 °C and 500–570 °C. The C component is a characteristic remanent magnetization (ChRM) and was isolated at higher temperatures, between 590–610 °C and 650–670 °C. The B component has been determined from 59 specimens distributed over 13 sites (Table 1): $N=13$, $D=138.0°$, $I=9.2°$, $k=388$, $\alpha_{95}=2.1°$ and $D=138.2°$, $I=13.1°$, $k=413$, $\alpha_{95}=2.0°$, before and after dip correction, respectively (Fig. 6). The B directions are a little bit better grouped in stratigraphic coordinates. But progressive untilting shows that the best clustering is maximum ($k=392$) around 53% de-tilting. A McFadden (1990) fold test gives an indeterminate result. The Tauxe and Watson (1994) fold test argues rather for a synfolding magnetization: optimal untilting at 95% of confidence between −53 and 137% de-tilting and optimal untilting at 63% of confidence between 12 and 97% de-tilting for site mean directions; optimal untilting at 95% confidence between 12 and 127% de-tilting and optimal untilting at 63% of

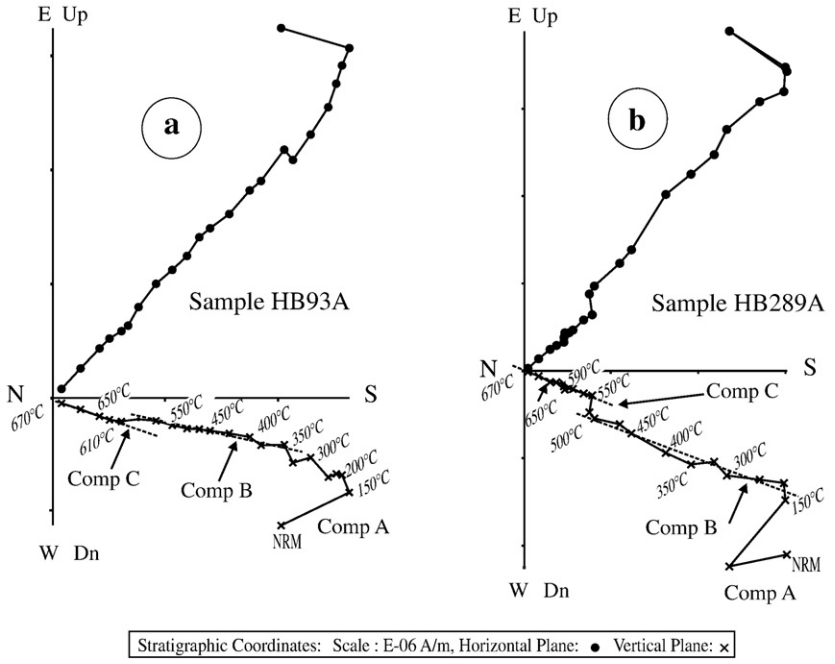


Fig. 5. Orthogonal vector plots for samples of the "Hassi Bachir" Formation (filled circles: horizontal plane, crosses: vertical plane).

Table 1
Mean direction (*D*, *I* in degrees) for B component obtained from each site (*N*: number of samples) and paleomagnetic pole: Lat (°S), Long (°E)

Before dip correction						After dip correction				Pole			
Site	*N*	*D* (°)	*I* (°)	*k*	α_{95}	*D* (°)	*I* (°)	*k*	α_{95}	Lat (°S)	Long (°E)	*K*	A_{95}
6	5	136.9	9.0	161	6.1	134.7	15.3	161	6.1				
7	5	141.6	5.8	143	6.4	140.0	12.9	143	6.4				
10	3	138.0	5.2	199	8.8	136.9	6.6	199	8.8				
22	5	133.7	10.2	197	5.5	136.6	15.4	176	15.4				
24	4	137.2	5.1	1747	2.2	138.3	11.1	1747	2.2				
25	5	137.0	9.4	547	3.3	137.4	12.6	547	3.3				
26	3	136.1	12.7	97	12.6	137.0	14.2	97	12.6				
27	5	134.5	16.4	265	4.7	134.9	19.1	265	4.7				
28	4	137.4	11.9	526	4.0	137.5	15.9	526	4.0				
30	5	141.4	7.9	579	3.2	141.3	11.8	579	3.2				
31	3	140.8	7.7	985	3.9	141.4	10.3	985	3.9				
32	5	139.4	7.0	65	9.5	140.2	9.0	65	9.5				
33	3	139.7	11.7	115	11.5	140.1	15.5	115	11.5				
Mean	59	137.3	9.6	180	1.4	137.4	13.5	194	1.3	36.7	59.0	–	–
Mean	13 sites	138.0	9.2	388	2.1	138.2	13.1	413	2.0	37.6	58.4	–	–
Mean	13 sites									37.6	58.4	822	1.4

Corresponding Fisher's (1953) parameters (*k*, α_{95}, *K*, A_{95}–α_{95} and A_{95} in degrees).

confidence between 47 and 102% de-tilting for individual samples direction (Fig. 7). Daly and Irving (1983) interpreted this component as a Permian chemical remagnetization. The paleomagnetic pole corresponding to this B component (for 53% de-tilting: 37.6°S, 60.0°E) is very close to their pole (35.5°S, 60.0°E). Bivariate statistics (Le Goff, 1990; Le Goff et al., 1992) indicate that the distribution of the directions B is significantly elongated before, during and after dip correction (*ky*/*kx* varying from 3.03 to 3.35 for a critical value — threshold of significant ellipticity — at 95% confidence of 2.59). The B component is therefore possibly a composite direction. This should be not surprising since ChRM C is probably also partly demagnetized for temperatures lower than 580 °C (Daly and Irving, 1983). For these temperatures, the unblocking temperature spectra of an unknown B' component (that should be the true Permian remagnetization) and of the C component should then be perfectly superimposed to yield linear segments on Zijderveld plots (Bouabdallah et al., 2003).

The ChRM C has been obtained with 69 specimens distributed over 15 sites (Table 2): $N=15$, $D=136.1°$, $I=18.0°$, $k=171$, $\alpha_{95}=2.9°$ and $D=136.1°$, $I=22.0°$, $k=217$, $\alpha_{95}=2.6°$ before and after dip correction, respectively (Fig. 8). The directions C are better grouped in stratigraphic coordinates. The distribution of ChRM C directions is only slightly lengthened as shown by the bivariate statistics (*ky*/*kx* between 1.8 and 2.2 for a critical value of 2.44). The *k* value during progressive untilting shows optimal clustering for about 90% de-tilting, but the corresponding *k* value (214.3) is not significantly different from that for 100% de-tilting (213.7). The McFadden (1990) fold test gives an indeterminate result for mean site directions ($\xi_{95}=4.510$, $\xi_{0\%}=4.090$, $\xi_{100\%}=2.108$), but clearly suggests pre-folding magnetization for samples directions ($\xi_{95}=9.732$, $\xi_{0\%}=24.647$, $\xi_{100\%}=6.643$ with optimal untilting since 86%). Similarly the fold test of Tauxe and Watson (1994) is almost indeterminate with regard to site mean direction (optimal untilting at 95% of confidence between 2 and 172% de-tilting and optimal untilting at 63% of confidence between 57 and 132% de-tilting) but much more significant with regard to samples directions (optimal untilting at 95% of confidence between 57 and 157% de-tilting and optimal untilting at 63% of confidence between 82 and 132% de-tilting — Fig. 7). Folding started during deposition (as shown by the slumping observed on the field), the presence of pre-folding magnetization is a clear proof of the primary character of the ChRM C.

The paleomagnetic pole corresponding to this ChRM C (32.8°S, 55.7°E, $K=328$ and $A_{95}=2.0°$) is slightly different from that (26.8°S, 56.6°E, $A_{95}=3.7°$) computed by Daly and Irving (1983), probably because their paleomagnetic pole was isolated using thermal demagnetization only up to 600 °C. This pole, which is more rigorously defined and better constrained in age than the pole of Daly and Irving (1983), is located in the vicinity of the other African Mid-Upper Carboniferous poles (Derder et al., 2001). Our pole has been obtained

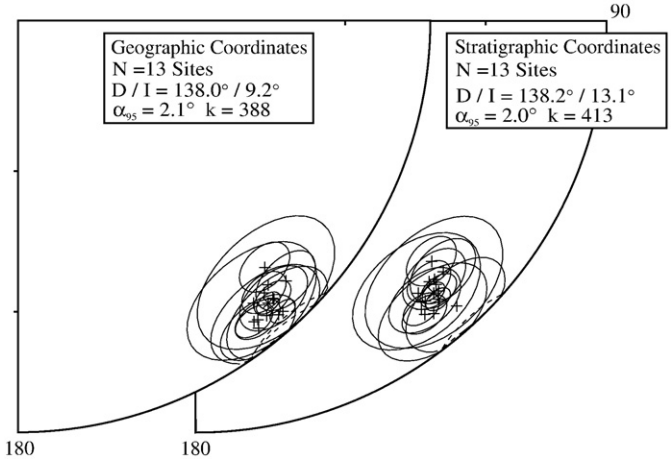


Fig. 6. "Hassi Bachir" Formation: Equal area plot (lower hemisphere) of the B component mean site directions; and their associated 95% confidence cone before and after dip correction.

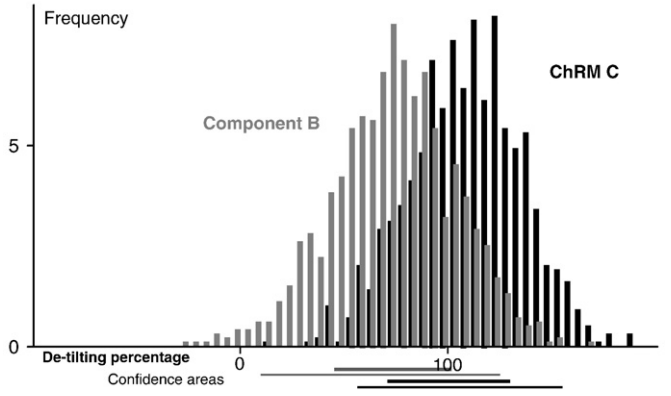


Fig. 7. Frequency (in percentage) of optimal untilting value (by window of 5°) obtained by the bootstrap method of Tauxe and Watson (1994) for B component (gray) and ChRM C (black), with confidence areas at 95% (thin line) and 63% (thick line).

Table 2
Mean direction (*D*, *I* in degrees) for C ChRM obtained from each site (*N*: number of samples) and paleomagnetic pole: Lat (°S), Long (°E)

Before dip correction						After dip correction				Pole			
Site	N	D (°)	I (°)	k	α_{95}	D (°)	I (°)	k	α_{95}	Lat(°S)	Long(°E)	K	A_{95}
5	7	144.7	11.5	118	5.6	142.2	19.0	118	5.6				
6	5	137.0	21.5	67	9.4	132.3	27.5	67	9.4				
7	4	138.5	15.8	389	4.7	135.0	17.1	389	4.7				
10	4	136.0	16.3	153	7.4	133.0	17.1	153	7.4				
19	4	129.8	16.6	188	6.7	130.3	19.2	188	6.7				
22	5	124.2	20.3	204	5.4	129.4	27.3	181	5.7				
24	5	132.3	10.5	757	2.8	134.2	17.1	757	2.8				
25	5	136.8	19.1	361	4.0	137.8	22.2	361	4.0				
26	4	136.7	18.8	148	7.6	138.0	20.3	148	7.6				
27	5	133.4	27.8	299	4.4	134.0	30.5	299	4.4				
28	4	136.9	20.9	490	4.2	137.1	24.9	490	4.2				
30	5	140.4	19.32	217	5.2	140.2	23.3	217	5.2				
31	4	139.8	14.1	189	6.7	141.1	16.6	189	6.7				
32	5	136.4	17.1	49	11.1	138.2	19.4	49	11.1				
33	3	138.3	19.1	291	7.2	138.8	22.9	291	7.2				
Mean	69	135.8	18.4	99	1.7	135.6	22.6	120	1.6	32.2	55.9	–	–
Mean	15 sites	136.1	18.0	171	2.9	136.1	22.0	217	2.6	32.8	55.8	–	–
Mean	15 sites									32.8	55.7	328	2.0

Corresponding Fisher's (1953) parameters (*k*, α_{95}, *K*, A_{95}–α_{95} and A_{95} in degrees).

from more samples and more sites distributed all over the structural basin and with more complete demagnetization than that obtained by Daly and Irving (1983). It therefore improves the quality of the African APWP.

5.2. The "Djebel Berga" Formation

In the late Visean–lower Namurian levels, the magnetic behavior during thermal demagnetization allows isolation of a ChRM, besides a viscous component A. This ChRM was generally defined between 200–250 °C and 500–570 °C. At higher temperatures the magnetic directions are highly disturbed (Fig. 9a), except for a few specimens (Fig. 9b and c), where the ChRM is isolated between 250 and 650°. According to the samples, this ChRM has two main directions corresponding to two magnetic components ("b" and "c").

On equal area plots, the obtained ChRM directions present different clusters (Fig. 10). ChRM directions "b" (N=3, D=142.8°, I=5.8°, k=44, α_{95}=18.8° and D=145.9°, I=10.4°, k=44, α_{95}=18.8°, before and after dip correction, respectively) have been found in 3 sites (13, 17 and 23). This component is close to the component B of the "Hassi Bachir" Formation. Three sites (13, 16 and 17) give other ChRM directions ("c") with higher inclination (D=124.0°, I=21.9°, k=1597, α_{95}=3.1° and D=134.3°, I=33.3°, k=419, α_{95}=6.0°, before and after dip correction, respectively). Finally, ChRM in site 18 has an intermediate inclination between those of the two main directions "b" and "c" (D=124.6.1°, I=13.5°, k=158, α_{95}=4.8° and D=128.6°, I=21.6°, k=158, α_{95}=4.8°, before and after dip correction, respectively). For these directions "b" and "c", fold tests are not conclusive. Clustering appears to be better before dip correction for the "b" component, but that cannot be considered as significant since this result corresponds only to 3 sites.

Contrary to the case of the components B and C of the "Hassi Bachir" Formation, directions "b" and "c" are not associated with different windows of temperature during the thermal analysis of the NRM. No difference appears when one compares magnetization behaviors on a Zijderveld plot. Rocks are moreover quite homogeneous in the "Djebel Berga" Formation sites. Therefore, no actual arguments except the difference in direction allow us to differentiate samples with directions "b" and "c". Both directions have been observed even in a same site (sites 13 and 17). Although well-defined data have been mostly obtained on Zijderveld plots for the "Djebel Berga" Formation, it remains then possible that these results correspond to composite magnetizations (Bouabdallah et al., 2003). For this reason and because of the low number of corresponding sites, directions "b" and "c" cannot be considered as reliable.

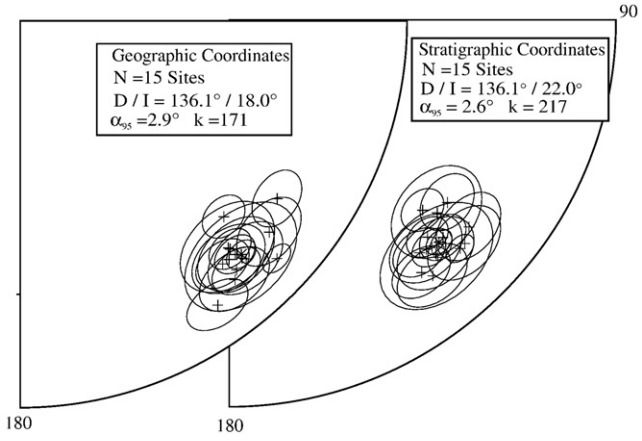


Fig. 8. "Hassi Bachir" Formation: Equal area plot (lower hemisphere) of the ChRM C mean site directions; and their associated 95% confidence cone before and after dip correction.

6. Structural implications

Only results from the "Hassi Bachir" Formation will therefore be retained. Fig. 7 shows that for the same untilting, the results of the Tauxe and Watson (1994) test are different for components B and C, and therefore that these components were not acquired at a same degree of tilting. According to the progressive untilting, composite component B should be "synfolding" (for about 53% de-tilting). The Tauxe and Watson (1994) fold test for samples directions also indicates that this component B is not pre-folding (mode of the distribution on Fig. 7: 75% de-tilting). Part of the folding within the structural basin of "Djebel Berga" is synsedimentary, and component B has been then, at least partly, acquired after this first folding.

To explain the optimal 50 to 75% de-tilting for component B, two extreme possibilities can be proposed, taking into account that the composite component B results from superimposition of a remagnetization B' and of the ChRM C. (i) Remagnetization B' and pre-folding ChRM C represent neighboring proportions of the component B: Permian remagnetization B' is most likely post-folding and the

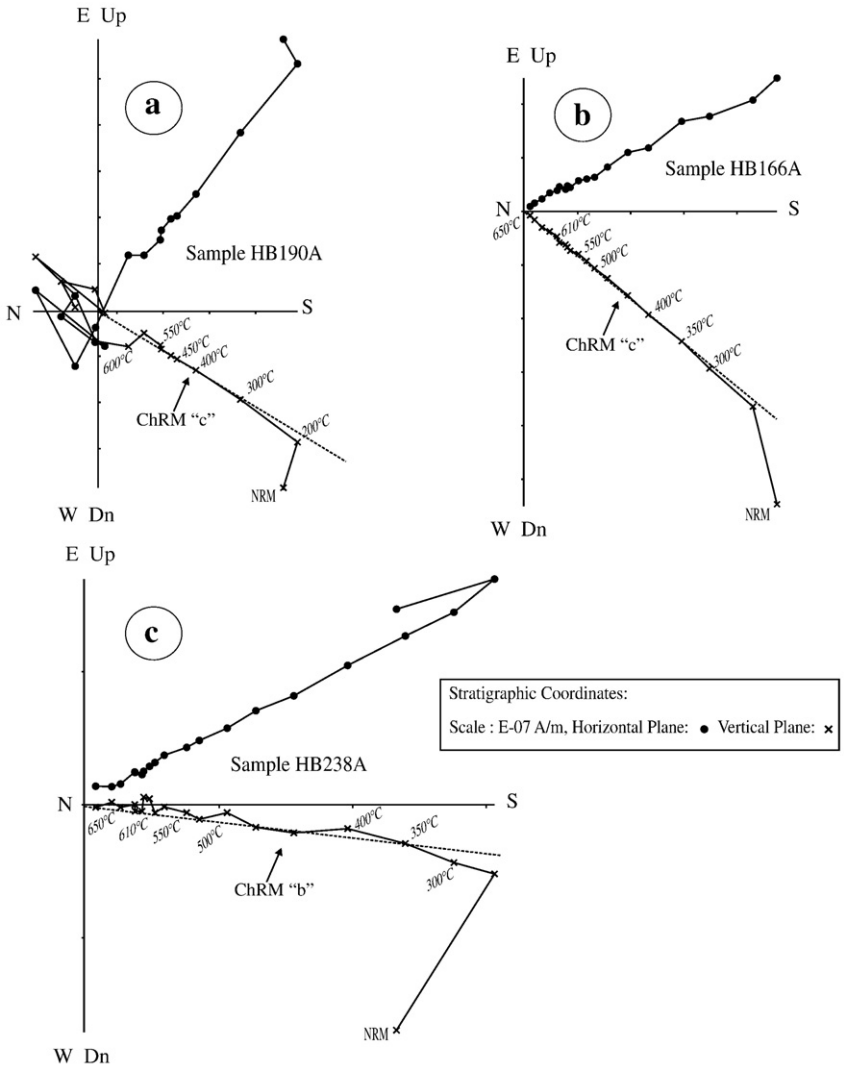


Fig. 9. Orthogonal vector plots for samples of the "Djebel Berga" Formation (filled circles: horizontal plane, crosses: vertical plane).

synfolding character of the B component is simply related to the superimposition of post-folding and pre-folding magnetizations. (ii) ChRM C is a weak part of the composite component B. Permian remagnetization B' is then actually synfolding and that implies a significant post-Permian folding. The resulting direction B and the component directions B' and C are included in a same plane. This

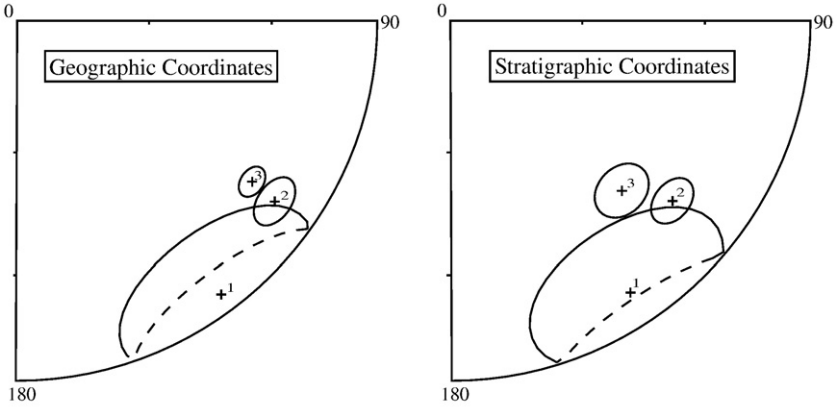


Fig. 10. "Djebel Berga" Formation: Equal area plot (lower-continuous line- and upper-dotted line-hemispheres) of the ChRM mean directions for components "b" (1), intermediate (2) and "c" (3); with their associated 95% confidence cone before and after dip correction.

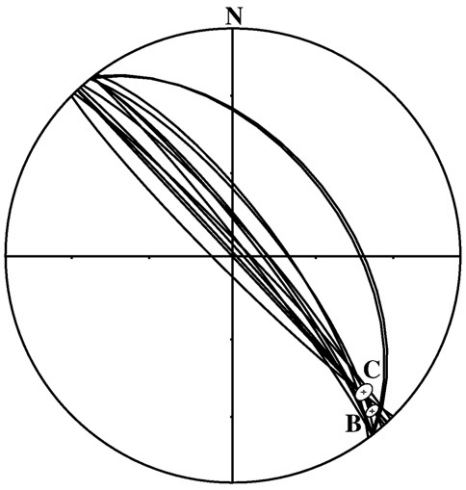


Fig. 11. "Hassi Bachir" Formation: Great circles including B and C mean directions for each site and mean B and C directions (with their associated 95% confidence cone) from all the sites (equal area plot in the lower hemisphere).

plane was determined for each site where directions B and C have been isolated (we present below the results before dip correction, but the obtained conclusion with bedding corrected data is exactly the same). The best intersection of the planes for the different sites using McFadden and McElhinny (1988) method (Fig. 11) then corresponds to the direction of the remagnetization B' (N=13, D=137.2°, I=8.6°, k=510, α_{95}=2.0°). This direction is not significantly different from the corresponding mean direction B (N=13, D=137.3°, I=9.2°, k=364, α_{95}=2.1°) but clearly differs from the ChRM C (N=15, D=136.1°, I=18.0°, k=171, α_{95}=2.9°). Part of ChRM C in the resulting component B is therefore weak and assumption (i) has to be rejected. Post-Permian tectonics are then responsible for about 50 to 75% total folding in the "Djebel Berga" basin, like 150 km to the West, in the Bled-el-Mass–Azzel Matti area in the "Reggane" basin (Smith et al., 2006). This work highlights the importance of post-Permian tectonics, likely a far-field stress effect of either the Cimmerian (~140 My) or the Austrian (~125 My) tectonic phase. Such a tectonic phase is probably a local consequence of a much larger scale major event related to the breakup of Gondwana in the late Barremian times, when South America started drifting away from Africa, while the rifts and fracture systems delimited three large blocks in the African plate.

7. Conclusions

Detailed analysis of paleomagnetic data during progressive untilting appears to be a powerful tool to reconstitute the structural evolution of a deformed area (e.g. Villalain et al., 2003; Henry et al., 2004b; Soto et al., 2008). Often none of the other classical structural approaches allows such a separation of the effects of different tectonic phases. The "Djebel Berga" structural basin is a very good example of superimposed tectonics. Syn-sedimentary deformation was followed not only by the well-known Hercynian folding but also by post-Permian tectonics. The latter represent a very significant part of the total folding and have to be taken into account in tectonic reconstructions of all the Saharan basins. This highlights again the interest of paleomagnetic studies in deformed areas, primary magnetization improving information about the Upper Carboniferous evolution of the whole African plate and remagnetization recording effects of previously underestimated Mesozoic regional tectonics.

Acknowledgements

This project is supported by the Algerian « PNR » program N° 04/04/13/99. This work benefited from careful and constructive reviews by G. Borradaile and an anonymous reviewer. We are very grateful to J. Boissonnas for the strong help with the manuscript, to the civil and military authorities at In Salah, and to the "Office du Parc National de l'Ahaggar" (OPNA) for their help.

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